

FoldAgent on BrowseComp-Plus: context efficiency vs end-to-end token usage

Build: `python3 docs/reports/build_e2e_report.py` (this file, assets and figures are regenerated from the HDFS ledgers). Repo commit at build time: 8aa019b.

1. Question and design

Does FoldAgent's folding reduce **active context** while increasing the total interaction needed to solve the same task? Three efficiencies are separated: **context efficiency** (peak / mean active context per forward pass), **trajectory efficiency** (turns, tool calls, tokens generated), and **end-to-end inference efficiency** (every token generated or returned before anything was folded away, and the cumulative model input over all forward passes).

Cells. Training $\in$ {no-RL Qwen3-8B base, GRPO checkpoint 60 (59 real steps), FoldGRPO checkpoint 50 (46 real steps)} $\times$ inference $\in$ {**no fold**: `plugin.workflow=search` (the branch tool is removed from the prompt; single-thread ReAct with `search/open_page/finish`), **fold**: `plugin.workflow=search_branch` (the training-time prompt with the branch/return tools; the policy decides when to branch, a branch inherits the main context, runs its own sub-trajectory and is folded to its return message)}. Both RL checkpoints were trained with the branch tool available: "vanilla GRPO" is the authors' baseline (same agent, plain GRPO, no fold-specific process reward), not a branch-free ReAct-GRPO. The no-fold cells are therefore an inference-time ablation for every model.

Evaluation. Identical to the fix-campaign final evals: 150-item BC-Plus test split, two decodings — greedy ($n=1$) and $T=1.0$ top- $p=1.0$ $n=4$ — prompt 8k + response window 32k per thread, `max_turn` 100 across threads, up to 10 branches, gpt-oss-120b judge, self-contained $8 \times H100$ val-only batch jobs (`infra/worker_val_selfcontained.sh`, `FOLD_WORKFLOW`, `FOLD_E2E_LEDGER=1`). Same tasks and same sampling settings across cells; task-level pairing by `instance_id` (t1n4 pairs the per-task mean of 4 samples).

Ledger. `agents/e2e_ledger.py` records, for every thread of every rollout, each turn's exact token ids the policy generated or was fed (`scripts/e2e_metrics.py` defines every metric). Folded-away branch tokens are counted in full; a branch's inherited context is counted as model input of its forward passes, not as new tokens.

cell	mode	results dir tag	rollouts	status	trainer val score
base_nofold	greedy	norl_base_greedy_e2e_nobbranch_sc	150	DONE	0.087
base_nofold	t1n4	norl_base_t1n4_e2e_nobbranch_sc	600	DONE	0.073
base_fold	greedy	norl_base_greedy_e2e_branch_sc	150	DONE	0.147
base_fold	t1n4	norl_base_t1n4_e2e_branch_sc	600	DONE	0.180
grpo_nofold	greedy	grpo_60_greedy_e2e_nobbranch_sc	150	DONE	0.153
grpo_nofold	t1n4	grpo_60_t1n4_e2e_nobbranch_sc	600	DONE	0.182
grpo_fold	greedy	grpo_60_greedy_e2e_branch_sc	150	DONE	0.327
grpo_fold	t1n4	grpo_60_t1n4_e2e_branch_t4h_sc	600	DONE	0.340
fold_nofold	greedy	foldgrpo_50_greedy_e2e_nobbranch_sc	150	DONE	0.067
fold_nofold	t1n4	foldgrpo_50_t1n4_e2e_nobbranch_sc	600	DONE	0.053
fold_fold	greedy	foldgrpo_50_greedy_e2e_branch_sc	150	DONE	0.373
fold_fold	t1n4	foldgrpo_50_t1n4_e2e_branch_t4h_sc	600	DONE	0.367

2. Main tables

2.1 All rollouts — greedy

tokens:

cell (train / inference)	n	success	e2e tokens	e2e median	generated	observations	fold in	fold gen	cum. input
base / no fold	150	0.087	36,478	39,906	4,272	27,040	0	0	117,601
base / fold	150	0.147	25,545	20,538	4,581	14,594	1,374	3,199	103,720
GRPO / no fold	150	0.153	36,754	39,044	3,159	28,430	0	0	110,043
GRPO / fold	150	0.327	62,627	45,094	4,301	51,531	1,755	2,872	379,794
FoldGRPO / no fold	150	0.067	39,315	39,814	2,064	32,082	0	0	150,772
FoldGRPO / fold	150	0.373	101,087	80,578	4,311	88,538	3,154	3,384	595,451

interaction and context:

cell (train / inference)	n	turns	tool calls	branches	fwd passes	peak ctx	mean ctx	main growth	overlong	post-fold tokens	post-fold turns	post-fold tool calls
base / no fold	150	5.6	5.3	0.00	5.6	31,778	19,033	31,329	0.76	-	-	-
base / fold	150	7.3	2.6	1.72	7.3	20,213	12,276	9,394	0.11	8,678	4.4	1.1
GRPO / no fold	150	5.4	5.0	0.00	5.4	30,734	19,335	31,605	0.71	-	-	-
GRPO / fold	150	19.5	12.0	2.43	19.5	31,135	18,293	10,832	0.11	38,554	14.4	8.3
FoldGRPO / no fold	150	6.8	6.7	0.00	6.8	33,871	21,637	34,166	0.91	-	-	-
FoldGRPO / fold	150	31.4	18.6	4.43	31.4	32,426	17,343	4,894	0.03	73,970	25.0	14.6

2.2 All rollouts — T=1.0, n=4

tokens:

cell (train / inference)	n	success	e2e tokens	e2e median	generated	observations	fold in	fold gen	cum. input
base / no fold	600	0.073	37,334	39,758	3,526	28,641	0	0	133,625
base / fold	600	0.180	25,703	21,678	4,685	14,647	1,374	3,272	109,332
GRPO / no fold	600	0.182	43,786	39,118	3,312	35,309	0	0	112,726
GRPO / fold	600	0.340	67,154	50,990	4,783	55,414	1,909	3,185	436,039
FoldGRPO / no fold	600	0.053	39,117	40,035	1,972	31,975	0	0	152,424
FoldGRPO / fold	600	0.367	99,818	84,308	4,342	87,296	3,095	3,362	615,312

interaction and context:

cell (train / inference)	n	turns	tool calls	branches	fwd passes	peak ctx	mean ctx	main growth	overlong	post-fold tokens	post-fold turns	post-fold tool calls
base / no fold	600	6.2	6.0	0.00	6.2	32,517	20,020	32,185	0.82	-	-	-
base / fold	600	7.7	2.5	1.75	7.7	20,170	11,977	8,692	0.08	8,631	4.6	1.1
GRPO / no fold	600	5.5	5.1	0.00	5.5	31,155	19,510	38,638	0.67	-	-	-
GRPO / fold	600	21.7	13.3	2.63	21.7	31,572	18,807	10,683	0.09	41,183	15.6	9.1

cell (train / inference)	n	turns	tool calls	branches	fwd passes	peak ctx	mean ctx	main growth	overlong	post-fold tokens	post-fold turns	post-fold tool calls
FoldGRPO / no fold	600	7.0	6.9	0.00	7.0	33,864	21,238	33,968	0.91	-	-	-
FoldGRPO / fold	600	32.2	19.5	4.48	32.2	32,568	17,563	5,009	0.03	71,346	25.2	14.9

2.3 Successful rollouts only — greedy

tokens:

cell (train / inference)	n	success	e2e tokens	e2e median	generated	observations	fold in	fold gen	cum. input
base / no fold	13	1.000	22,031	20,648	2,619	14,249	0	0	60,636
base / fold	22	1.000	20,096	18,512	3,412	10,830	864	1,862	78,467
GRPO / no fold	23	1.000	23,118	22,492	2,336	15,617	0	0	79,277
GRPO / fold	49	1.000	38,621	29,233	2,921	29,515	1,171	1,917	237,970
FoldGRPO / no fold	10	1.000	23,426	24,462	1,587	16,669	0	0	92,984
FoldGRPO / fold	56	1.000	58,653	48,424	3,086	48,343	2,184	2,430	330,826

interaction and context:

cell (train / inference)	n	turns	tool calls	branches	fwd passes	peak ctx	mean ctx	main growth	overlong	post-fold tokens	post-fold turns	post-fold tool calls
base / no fold	13	4.1	2.9	0.00	4.1	22,031	14,410	16,880	0.00	-	-	-
base / fold	22	5.9	2.3	1.09	5.9	18,626	11,818	7,807	0.00	2,466	1.9	0.2
GRPO / no fold	23	4.7	3.6	0.00	4.7	23,118	16,208	17,966	0.00	-	-	-
GRPO / fold	49	13.8	8.6	1.59	13.8	26,864	15,600	6,593	0.00	15,682	7.7	4.3
FoldGRPO / no fold	10	5.7	4.6	0.00	5.7	23,426	15,839	18,274	0.00	-	-	-
FoldGRPO / fold	56	20.6	12.0	3.07	20.6	27,852	14,494	2,754	0.00	33,427	14.1	7.8

2.4 Successful rollouts only — T=1.0, n=4

tokens:

cell (train / inference)	n	success	e2e tokens	e2e median	generated	observations	fold in	fold gen	cum. input
base / no fold	44	1.000	22,177	22,110	2,141	14,871	0	0	86,203
base / fold	108	1.000	23,787	21,483	3,380	14,350	1,066	1,923	86,989
GRPO / no fold	109	1.000	23,650	23,772	2,259	16,232	0	0	81,853
GRPO / fold	204	1.000	41,605	33,226	3,164	32,082	1,337	2,079	262,765
FoldGRPO / no fold	32	1.000	23,408	24,499	1,498	16,753	0	0	94,055
FoldGRPO / fold	220	1.000	58,582	47,287	3,205	48,125	2,210	2,511	346,632

interaction and context:

cell (train / inference)	n	turns	tool calls	branches	fwd passes	peak ctx	mean ctx	main growth	overlong	post-fold tokens	post-fold turns	post-fold tool calls
base / no fold	44	5.3	4.2	0.00	5.3	22,177	15,321	17,029	0.00	-	-	-
base / fold	108	6.7	2.8	1.30	6.7	20,383	12,018	7,255	0.00	5,926	3.2	0.9
GRPO / no fold	109	4.9	3.7	0.00	4.9	23,650	16,178	18,505	0.00	-	-	-
GRPO / fold	204	14.9	9.1	1.81	14.9	27,336	16,209	6,989	0.00	17,326	8.1	4.4
FoldGRPO / no fold	32	5.9	4.9	0.00	5.9	23,408	15,678	18,268	0.00	-	-	-
FoldGRPO / fold	220	21.3	12.7	3.10	21.3	28,242	14,714	3,139	0.00	33,851	14.5	8.2

Columns: e2e tokens = prompt + generated + framing + observations + fold prompts/returns (everything before discard); fold in = branch prompts, overflow-summary prompts and folded return messages injected into contexts; fold gen = policy tokens spent on branch-call / return / summary turns; cum. input = Σ input length over all forward passes; peak ctx = max input+output length of any forward pass; post-fold = after the first fold (rollouts that folded; NaN otherwise). Means with bootstrap CIs are in assets/cell_summary.csv.

2.5 Termination

Greedy:

cell	n	finish	max_turn	session timeout	window exhausted	overlong (main $\geq 32k$)	folded ≥ 1	branch limit
base / no fold	150	24%	0%	0%	76%	76%	0%	0%
base / fold	150	89%	0%	0%	11%	11%	87%	0%
GRPO / no fold	150	29%	0%	0%	71%	71%	0%	0%
GRPO / fold	150	89%	0%	0%	11%	11%	85%	1%
FoldGRPO / no fold	150	8%	0%	0%	92%	91%	0%	0%
FoldGRPO / fold	150	97%	0%	0%	3%	3%	98%	13%

T=1.0 n=4:

cell	n	finish	max_turn	session timeout	window exhausted	overlong (main $\geq 32k$)	folded ≥ 1	branch limit
base / no fold	600	18%	0%	0%	82%	82%	0%	0%
base / fold	600	92%	0%	0%	8%	8%	87%	0%
GRPO / no fold	600	33%	0%	0%	67%	67%	0%	0%
GRPO / fold	600	91%	0%	0%	9%	9%	90%	2%
FoldGRPO / no fold	600	9%	0%	0%	91%	91%	0%	0%
FoldGRPO / fold	600	96%	0%	0%	3%	3%	99%	9%

3. Paired task-level comparisons

Greedy (one rollout per task):

comparison	metric	n tasks	mean Δ [95% CI]	median Δ	% tasks $\Delta > 0$	sign-test p
FoldGRPO/fold – GRPO/fold (same tasks)	success	150	0.047 [-0.027, 0.120]	0.000	13%	0.281

comparison	metric	n tasks	mean Δ [95% CI]	median Δ	% tasks $\Delta > 0$	sign-test p
FoldGRPO/fold – GRPO/fold (same tasks)	total_e2e_tokens	150	38,459 [26,383, 51,096]	20,903	69%	0.000
FoldGRPO/fold – GRPO/fold (same tasks)	turns	150	11.9 [7.9, 15.9]	9.0	69%	0.000
FoldGRPO/fold – GRPO/fold (same tasks)	tool_calls	150	6.6 [4.0, 9.2]	5.0	64%	0.000
FoldGRPO/fold – GRPO/fold (same tasks)	peak_ctx	150	1,291 [-50, 2,709]	322	54%	0.369
FoldGRPO/fold – GRPO/fold (same tasks)	cum_prompt_tokens	150	215,657 [128,254, 301,375]	138,373	63%	0.001
FoldGRPO: fold – no fold	success	150	0.307 [0.227, 0.387]	0.000	33%	0.000
FoldGRPO: fold – no fold	total_e2e_tokens	150	61,772 [51,125, 72,872]	38,906	83%	0.000
FoldGRPO: fold – no fold	turns	150	24.6 [21.3, 28.1]	19.0	90%	0.000
FoldGRPO: fold – no fold	tool_calls	150	11.9 [9.7, 14.0]	8.5	81%	0.000
FoldGRPO: fold – no fold	peak_ctx	150	-1,444 [-2,741, -147]	234	54%	0.326
FoldGRPO: fold – no fold	cum_prompt_tokens	150	444,679 [371,266, 527,580]	295,167	85%	0.000
GRPO: fold – no fold	success	150	0.173 [0.100, 0.247]	0.000	21%	0.000
GRPO: fold – no fold	total_e2e_tokens	150	25,873 [18,490, 33,792]	8,762	65%	0.000
GRPO: fold – no fold	turns	150	14.1 [11.5, 16.9]	9.5	81%	0.000
GRPO: fold – no fold	tool_calls	150	6.9 [5.2, 8.8]	4.0	66%	0.000
GRPO: fold – no fold	peak_ctx	150	401 [-1,070, 1,924]	766	52%	0.683
GRPO: fold – no fold	cum_prompt_tokens	150	269,751 [212,807, 328,598]	168,095	76%	0.000
FoldGRPO/no fold – GRPO/no fold	success	150	-0.087 [-0.147, -0.027]	0.000	3%	0.011
FoldGRPO/no fold – GRPO/no fold	total_e2e_tokens	150	2,561 [705, 4,364]	630	57%	0.086
FoldGRPO/no fold – GRPO/no fold	turns	150	1.4 [0.9, 2.0]	2.0	68%	0.000
FoldGRPO/no fold – GRPO/no fold	tool_calls	150	1.7 [1.2, 2.3]	2.0	69%	0.000
FoldGRPO/no fold – GRPO/no fold	peak_ctx	150	3,136 [1,867, 4,442]	1,714	61%	0.009
FoldGRPO/no fold – GRPO/no fold	cum_prompt_tokens	150	40,728 [26,814, 54,441]	46,297	73%	0.000
base: fold – no fold	success	150	0.060 [-0.013, 0.133]	0.000	13%	0.136
base: fold – no fold	total_e2e_tokens	150	-10,933 [-13,825, -8,125]	-13,123	27%	0.000
base: fold – no fold	turns	150	1.7 [0.8, 2.6]	1.0	55%	0.012
base: fold – no fold	tool_calls	150	-2.7 [-3.5, -2.0]	-3.0	21%	0.000
base: fold – no fold	peak_ctx	150	-11,566 [-13,605, -9,500]	-14,278	23%	0.000
base: fold – no fold	cum_prompt_tokens	150	-13,880 [-34,924, 8,480]	-24,714	37%	0.001
FoldGRPO/fold – GRPO/no fold	success	150	0.220 [0.140, 0.300]	0.000	26%	0.000
FoldGRPO/fold – GRPO/no fold	total_e2e_tokens	150	64,332 [53,558, 75,568]	43,157	88%	0.000
FoldGRPO/fold – GRPO/no fold	turns	150	26.0 [22.6, 29.5]	21.0	97%	0.000
	tool_calls	150	13.6 [11.5, 15.7]	10.0	89%	0.000

comparison	metric	n tasks	mean Δ [95% CI]	median Δ	% tasks $\Delta > 0$	sign-test p
FoldGRPO/fold – GRPO/no fold						
FoldGRPO/fold – GRPO/no fold	peak_ctx	150	1,692 [207, 3,098]	1,466	63%	0.001
FoldGRPO/fold – GRPO/no fold	cum_prompt_tokens	150	485,407 [410,751, 562,530]	334,877	94%	0.000

T=1.0 n=4 (per-task mean of 4 samples):

comparison	metric	n tasks	mean Δ [95% CI]	median Δ	% tasks $\Delta > 0$	sign-test p
FoldGRPO/fold – GRPO/fold (same tasks)	success	150	0.027 [-0.008, 0.062]	0.000	20%	0.262
FoldGRPO/fold – GRPO/fold (same tasks)	total_e2e_tokens	150	32,663 [26,292, 39,001]	24,930	81%	0.000
FoldGRPO/fold – GRPO/fold (same tasks)	turns	150	10.5 [8.4, 12.6]	9.5	79%	0.000
FoldGRPO/fold – GRPO/fold (same tasks)	tool_calls	150	6.3 [4.9, 7.6]	5.4	79%	0.000
FoldGRPO/fold – GRPO/fold (same tasks)	peak_ctx	150	996 [393, 1,624]	648	57%	0.086
FoldGRPO/fold – GRPO/fold (same tasks)	cum_prompt_tokens	150	179,274 [133,413, 227,503]	141,262	78%	0.000
FoldGRPO: fold – no fold	success	150	0.313 [0.253, 0.377]	0.000	47%	0.000
FoldGRPO: fold – no fold	total_e2e_tokens	150	60,700 [52,755, 68,772]	47,522	92%	0.000
FoldGRPO: fold – no fold	turns	150	25.1 [22.6, 27.7]	21.1	100%	0.000
FoldGRPO: fold – no fold	tool_calls	150	12.6 [10.9, 14.3]	11.0	90%	0.000
FoldGRPO: fold – no fold	peak_ctx	150	-1,296 [-2,123, -541]	67	51%	0.807
FoldGRPO: fold – no fold	cum_prompt_tokens	150	462,888 [405,310, 522,060]	359,591	93%	0.000
GRPO: fold – no fold	success	150	0.158 [0.115, 0.203]	0.000	35%	0.000
GRPO: fold – no fold	total_e2e_tokens	150	23,368 [5,770, 34,364]	22,890	84%	0.000
GRPO: fold – no fold	turns	150	16.2 [14.4, 18.0]	14.0	99%	0.000
GRPO: fold – no fold	tool_calls	150	8.2 [7.0, 9.4]	6.9	90%	0.000
GRPO: fold – no fold	peak_ctx	150	417 [-400, 1,208]	983	57%	0.086
GRPO: fold – no fold	cum_prompt_tokens	150	323,313 [283,797, 365,376]	279,956	95%	0.000
FoldGRPO/no fold – GRPO/no fold	success	150	-0.128 [-0.168, -0.090]	0.000	2%	0.000
FoldGRPO/no fold – GRPO/no fold	total_e2e_tokens	150	-4,669 [-19,650, 3,296]	1,479	66%	0.000
FoldGRPO/no fold – GRPO/no fold	turns	150	1.5 [1.2, 1.9]	1.5	76%	0.000
FoldGRPO/no fold – GRPO/no fold	tool_calls	150	1.9 [1.5, 2.2]	1.8	81%	0.000
FoldGRPO/no fold – GRPO/no fold	peak_ctx	150	2,709 [2,043, 3,400]	2,070	72%	0.000
FoldGRPO/no fold – GRPO/no fold	cum_prompt_tokens	150	39,698 [30,450, 49,278]	35,758	79%	0.000
base: fold – no fold	success	150	0.107 [0.070, 0.147]	0.000	26%	0.000
base: fold – no fold	total_e2e_tokens	150	-11,631 [-13,076, -10,203]	-12,720	10%	0.000

comparison	metric	n tasks	mean Δ [95% CI]	median Δ	% tasks $\Delta > 0$	sign-test p
base: fold – no fold	turns	150	1.5 [0.7, 2.3]	0.5	59%	0.021
base: fold – no fold	tool_calls	150	-3.4 [-3.8, -3.1]	-3.5	7%	0.000
base: fold – no fold	peak_ctx	150	-12,347 [-13,224, -11,448]	-13,014	1%	0.000
base: fold – no fold	cum_prompt_tokens	150	-24,293 [-39,808, -6,712]	-44,094	28%	0.000
FoldGRPO/fold – GRPO/no fold	success	150	0.185 [0.137, 0.235]	0.000	37%	0.000
FoldGRPO/fold – GRPO/no fold	total_e2e_tokens	150	56,031 [37,115, 69,161]	50,009	96%	0.000
FoldGRPO/fold – GRPO/no fold	turns	150	26.7 [24.2, 29.3]	22.2	99%	0.000
FoldGRPO/fold – GRPO/no fold	tool_calls	150	14.4 [12.9, 16.0]	12.0	98%	0.000
FoldGRPO/fold – GRPO/no fold	peak_ctx	150	1,412 [677, 2,169]	2,058	69%	0.000
FoldGRPO/fold – GRPO/no fold	cum_prompt_tokens	150	502,587 [447,143, 559,534]	398,808	99%	0.000

4. Figures

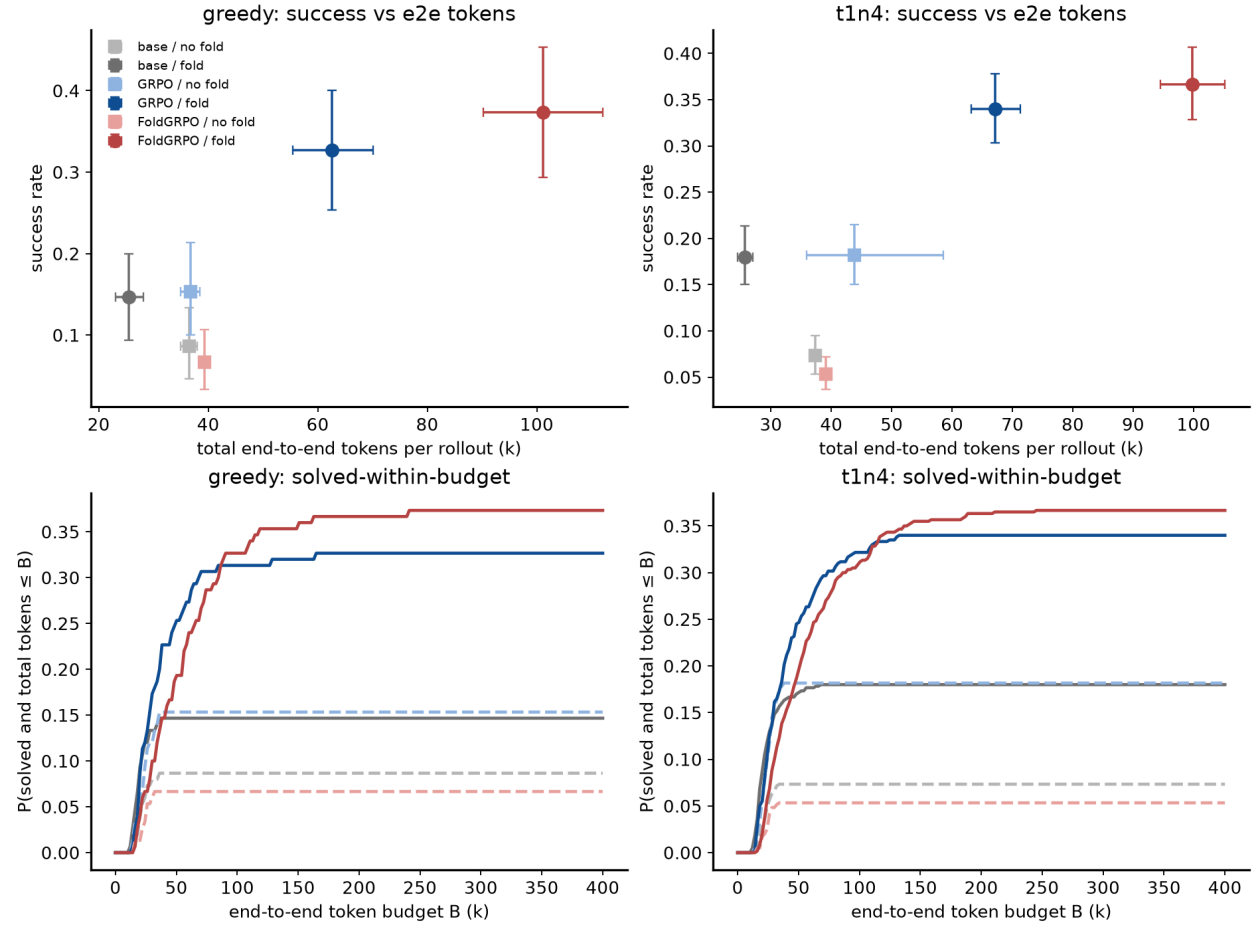
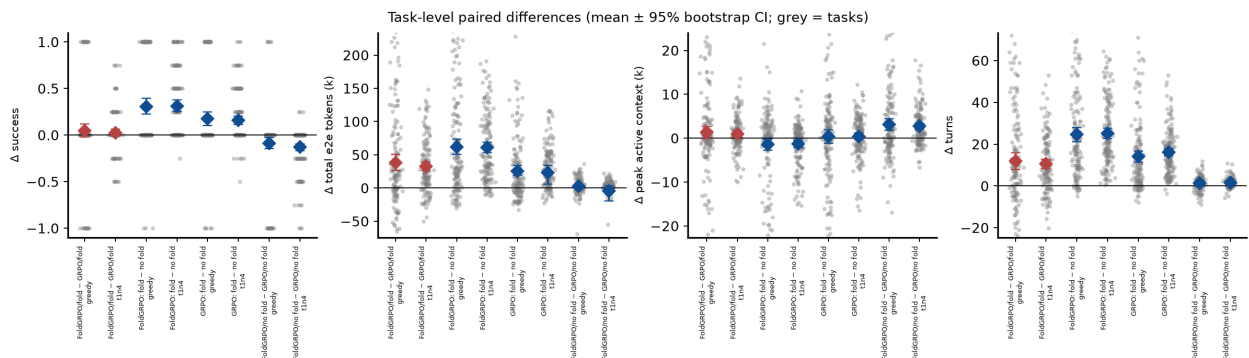
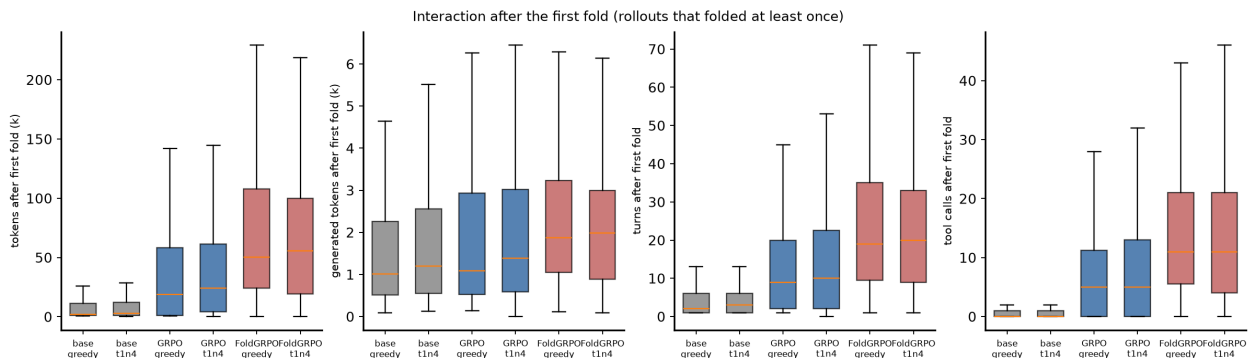
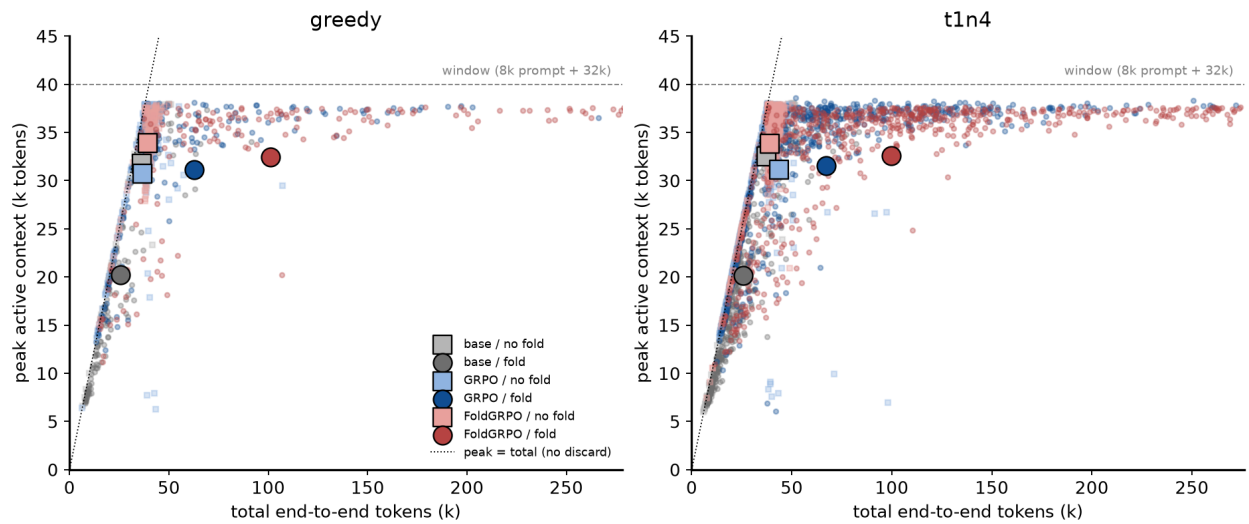


Fig. 1 — Success vs total end-to-end tokens (top: cell means $\pm$ 95% CI; bottom: fraction of rollouts solved within an end-to-end token budget).



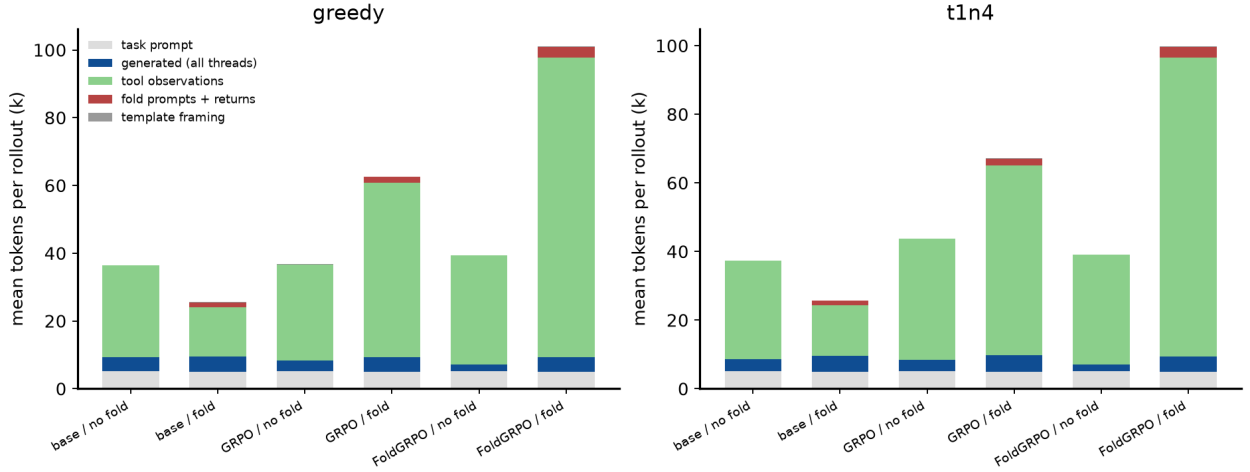


Fig. 5 — Where the end-to-end tokens go.

5. Findings

Headline. On the same 150 tasks, FoldAgent's folding does **not** buy shorter active context for free. For both RL checkpoints the peak active context is the same with or without folding (every policy saturates the 32k window somewhere in the trajectory; paired $\Delta_{\text{peak}} \approx 0$), the mean context per forward pass drops only 10–20 %, and in exchange the trajectory becomes 3–5× longer in turns and 1.7–2.6× longer in end-to-end tokens, with 3.5–5.5× the cumulative model input. Most of that extra interaction happens **after the first fold**. FoldGRPO training does not learn to compensate for this: it learns to fold more and to spend ~60 % more end-to-end tokens than the plain-GRPO policy on the same tasks, for a success gain that is not significant (greedy +0.047 [−0.03, 0.12]; T=1.0 n=4: +0.027 [−0.01, 0.06]).

1. Context efficiency (what folding actually changes). Greedy, all rollouts: peak active context 31.1k (GRPO/fold) vs 30.7k (GRPO/no fold) and 32.4k (FoldGRPO/fold) vs 33.9k (FoldGRPO/no fold) — paired $\Delta_{\text{peak}} +0.4\text{k}$ [−1.3k, 1.9k] and −1.4k [−2.8k, −0.2k]. The window is hit either in the main thread (no fold) or inside a branch (fold). Mean input length per forward pass: 18.3k / 17.3k with folding vs 19.3k / 21.6k without. Only the untrained base model gets a real context reduction from folding (peak 20.2k vs 31.8k, −11.6k paired), because it returns from branches after ~2 tool calls.

2. Trajectory efficiency (interaction). Folding multiplies interaction: turns 19.5 vs 5.4 (GRPO) and 31.4 vs 6.8 (FoldGRPO); tool calls 12.0 vs 5.0 and 18.6 vs 6.7; branches per rollout 2.4 (GRPO) and 4.4 (FoldGRPO). Of the fold cells' end-to-end tokens, 62 % (GRPO) and 73 % (FoldGRPO) are generated or returned after the first fold (38.6k / 74.0k tokens, 14.4 / 25.0 turns; Fig. 3). Restricting to successful rollouts the pattern is the same: a solved task costs 23.1k e2e tokens for GRPO/no fold, 38.6k for GRPO/fold and 58.7k for FoldGRPO/fold (20.1k for base/fold).

3. End-to-end inference efficiency. Greedy means (medians): e2e tokens 36.8k (39.0k) GRPO/no fold → 62.6k (45.1k) GRPO/fold → 101.1k (80.6k) FoldGRPO/fold; cumulative model input 110k → 380k → 595k tokens per rollout. Paired on the same tasks, FoldGRPO/fold − GRPO/fold = +38.5k e2e tokens [26k, 51k], +11.9 turns, +6.6 tool calls, +216k cumulative input (all $p < 0.001$), for +0.047 success ($p = 0.28$). The solved-within-budget curves (Fig. 1, bottom) show GRPO/fold dominating FoldGRPO/fold up to an end-to-end budget of ~100k tokens ($B = 50\text{k}$: 0.253 vs 0.193; 75k: 0.307 vs 0.287); FoldGRPO/fold only pulls ahead when $> 100\text{k}$ tokens per task are allowed (0.373 vs 0.327 at 400k).

4. Why the no-fold cells are not simply "cheaper". Without the branch tool both RL policies die at the window: 71 % (GRPO) and 91 % (FoldGRPO) of rollouts end with the 32k window exhausted before an answer (success 0.153 and 0.067; FoldGRPO/no fold is below the base

model, 0.087). Their low e2e token counts are the cost of a killed trajectory, not of an efficient one. Folding converts the hard window limit into a soft budget on interaction: finish rate 89 % / 97 %, window exhaustion 11 % / 3 %. So the honest reading is: folding buys survival past the window, and the trained policies pay for it with interaction, not with a smaller active context.

5. What FoldGRPO training learned. Relative to GRPO with the same tool: +2 branches per rollout, +60 % e2e tokens, 13 % of greedy rollouts hit the 10-branch limit (GRPO 1 %), and complete dependence on the tool (no-fold success 0.067 vs GRPO's 0.153). It did learn to stop: 97 % finish and 3 % window exhaustion (GRPO 89 % / 11 %), and its main thread is shorter — but the stopping is achieved by delegating more work to branches, i.e. by spending more end-to-end tokens, not by needing less post-fold interaction. Neither "reduces context occupancy without increasing future interaction" nor "learns to compensate for post-compaction interaction" holds; "trades shorter (mean) context for more search/reasoning" is the description the data supports.

6. Sampling (T=1.0, n=4) reproduces the greedy picture. With the fold cells re-run under a 4 h session timeout (see below), the sampled results agree with greedy on every axis: success 0.340 (GRPO/fold) vs 0.367 (FoldGRPO/fold), paired $\Delta = +0.027$ [$-0.008, 0.062$], $p = 0.26$; e2e tokens 67.2k vs 99.8k (paired +32.7k [26k, 39k]); turns 21.7 vs 32.2 (+10.5); branches 2.6 vs 4.5; cumulative model input 391k vs 574k (+179k); peak context 31.6k vs 32.6k ($\Delta +1.0$ k); finish rate 91 % vs 96 %, window exhaustion 9 % vs 3 %; 58 % vs 71 % of the fold cells' tokens fall after the first fold; a solved task costs 41.6k vs 58.6k e2e tokens. The solved-within-budget curves cross between 100k and 150k tokens per task (B = 50k: 0.247 vs 0.197; 100k: 0.322 vs 0.310; 150k: 0.340 vs 0.355; 400k: 0.340 vs 0.367).

Timeout artefact and re-runs. The val worker's default 1 h session timeout is not binding for greedy (median session 8–12 min) or for the no-fold cells, but the 600 concurrent sampled rollouts of the fold cells are throughput-bound on the shared search + judge pod (median session 50–55 min): the first T=1 runs lost 10 % (GRPO/fold) and 34 % (FoldGRPO/fold) of rollouts to the timeout, all scored 0 with truncated token counts (success read 0.317 / 0.317). Both cells were re-run with FOLD_SESSION_TIMEOUT=14400 (jobs c5220b294491facc and 5fc30e5bf85e8cc1; tag _e2e_branch_t4h; 0 timeouts in either) and the tables above use the re-runs. The 1 h results are kept under the original tags for reference; they underestimate both the success and the cost of folding.

6. Caveats

- Both RL checkpoints are truncated runs (46 / 59 real steps) from the fix campaign and were trained with the branch tool; the no-fold cells are an inference-time ablation with a different system prompt (search vs search_branch), so they measure "the same policy without the ability to fold", not a branch-free-trained baseline.
- "Compaction" here is FoldAgent's branch/fold; the CompactionRL-style summary compaction (compactiongrpo arm) was still training and is not included.
- Judge = gpt-oss-120b (not the paper's GPT models); success = judge-graded correctness of the final answer (0/1).
- The bottom of the 32k window: a thread that fills its window keeps its partial context; overlong marks main threads that reached the window.
- Token counts are exact ids from the ledger; the branch's inherited context is re-rendered by the chat template (± 1 token per inherited turn).
- Greedy decoding on vLLM is not bit-reproducible across nodes; t1n4 CIs reflect sampling noise over 4 samples $\times$ 150 tasks.

7. Reproduce

```
# 1. jobs (already submitted; specs in infra/jobs/fold_e2e*_h100.json, ledger in infra/jobs/JOBS.tsv)
cd ~/xiaoxuan/FoldAgent/infra/jobs && merlin-cli --control-plane i18n-tt job-v2 runs create --from-file fold_e2e_grpo60_branch_h100.json # ... x6
# 2. report (reads /mnt/hdfs/mlsys/xiaoxuan/fold_replication/results/valonly*_e2e*_sc/ledger)
python3 docs/reports/build_e2e_report.py
# tests: ~/xiaoxuan/envs/fold_train/bin/python -m unittest tests.test_e2e_ledger
tests.test_e2e_metrics
```

Artifacts: docs/reports/2026-09-17_foldagent_e2e_token_usage_report_assets/ (rollouts.csv = one row per rollout with every metric; cell_summary.csv; paired_diffs.csv; figures as PNG+PDF with JSON data).